MILITARY SPECIFICATION
INFLATION EQUIPMENT, CARBON DIOXIDE, FOR MULTIPLACE LIFE RAFTS

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for multiplace life rafts carbon dioxide inflation equipment with the remote pull type of actuation. The following life rafts are referred to throughout this specification:

LRU-1/P	LRU-14/A (Mark 12)
LRU-12/A (Mark 4)	LRU-15/A (Mark 20)
LRU-13/A (Mark 7)	F-2B

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-C-101	-	Carbon Dioxide (CO_2), Technical and USP.
BB-N-411	-	Nitrogen, Technical.
QQ-A-225/5	-	Aluminum Alloy 2017 Bar, Rod, and Wire; Rolled Drawn or Cold Finished.
MMM-A-139	-	Adhesive, Natural or Synthetic-Natural Rubber.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-T-60	-	Tape, Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-C-7905	-	Cylinders, Compressed Gas, Non-Shatterable.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-V-8883	-	Valve, Inlet, Life Raft.
MIL-C-23070	-	Cloth, Laminated, and Tape, Coated Cloth, Polyisoprene, Natural or Synthetic, Rubber on Nylon.

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services.
FED-STD-151	-	Metals, Test Methods.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-889	-	Dissimilar Metals.
MS26545	-	Cylinder - Compressed Gas, Non-Shatterable.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

63A120H1	-	Inflation Equipment, Carbon Dioxide, for Multiplace Life Rafts.
1106AS102	-	Housing Pull Cable, Inflation Equipment
1106AS103	-	Cable Assembly, Snaphook, Inflation Valve, Multi-Place Life Raft
1106AS104	-	Cable Assembly, Looped End, Inflation Valve Multi-Place Life Raft

AIR FORCE

55C3689	-	Cylinder and Valve Assembly, Carbon Dioxide, Multiplace Life Raft.

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PUBLICATIONS

DEPARTMENT OF TRANSPORATION

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Regulations for Transportation of Explosives and Other Dangerous Articles.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample of the carbon dioxide inflation equipment or component shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Materials and components. Materials and components shall conform to applicable specifications and drawings as listed or required herein. Materials and components which are not covered by applicable specifications or drawings, or are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and entirely suitable for the purpose intended. Unless otherwise specified, the materials and components, except for the metallic parts (excluding the carbon dioxide cylinder), used in the construction of the inflation equipment shall have been manufactured not more than 12 months prior to the date of delivery of the inflation equipment. No magnesium alloys shall be used.

3.2.1 Metals. The metals used in the inflation equipment shall be of the corrosion resistant type or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the inflation equipment may be subjected, when in storage or during normal service life.

3.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

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3.2.2 Protective treatment. When materials are used in the construction of the inflation equipment that are subject to deterioration, when exposed to environmental conditions likely to occur durng service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.3 Design and construction. The design and construction of the inflation equipment shall conform to Drawings 63A120H1 and 55C3689, as applicable.

3.3.1 Cylinder valve. The cylinder valve shall be designed to operate freely under all conditions to which it may be subjected and shall be operable from any direction of pull. The valve shall provide an unrestrictive passage of the carbon dioxide gas to the manifold without leakage. The valve shall operate and discharge the carbon dioxide gas through the manifold and into the raft, when the remote type of actuation is held and the raft is allowed to drop freely. Except for the actuation, the valve shall not have any parts easily removable by hand. The valve outlet shall be of the non-recoiling type and shall be designed to eliminate any recoil of the cylinder if discharged when being disconnected. A safety port shall be provided to release the carbon dioxide gas into the atmosphere in case of excessive cylinder pressure. The safety outlet shall be closed with a safety plug and frangible disc. The frangible disc shall be replaceable. The valve shall permit charging of the carbon dioxide cylinder through the outlet without removing the cylinder or replacement parts. The configuration of the valve assembly shall conform to Drawing 63A120H1-71.

3.3.2 Inlet manifold.

3.3.2.1 LRU-1/P, LRU-12/A, LRU-13/A, LRU-14/A life rafts. The inlet manifold for the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts shall consist of a metal body molded in a natural rubber case and mounted on a natural rubber patch. The manifold body shall be attached to one end of a flexible hose assembly. The other end of the flexible hose shall be attached to the cylinder valve. The manifold shall contain two check valves which shall permit the carbon dioxide gas to pass through the inlet manifold and to discharge equally through the inlet diffusers into the life raft tube. The inlet check valves shall prevent the carbon dioxide gas in the life raft tube from returning into the manifold. The manifold shall be designed that, when attached to a life raft, it shall readily permit the attachment and removal of a charged carbon dioxide cylinder and valve. The inlet manifold assembly for the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts shall conform to Drawing 63A120H1-30.

3.3.2.1.1 Inlet diffusers. The manifold shall contain two inlet diffusers and each shall be an integral part of the manifold body. Each diffuser shall have not less than three outlet holes which shall be drilled at an angle to the manifold base. The discharge holes shall be so located that recoil in any direction shall be kept at a minimum, when the carbon dioxide gas is discharged into the raft inflatable tube.

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3.3.2.1.2 Patch. A patch, fabricated from natural rubber, shall be securely molded to the body of the inlet manifold. The bonding material shall not affect the rubber patch or metal and shall not exude, become brittle, or stiff. The side of the patch, to be cemented to the outer side of the raft inflatable tube, shall be flat. The composition shall be such that it shall have good cementability properties and shall be capable of being cemented, without difficulty, to the life raft tube cloth conforming to MIL-C-23070, Variety C, with the adhesive conforming to MMM-A-139, Class 1 or 2. Except for the discharge holes, all the parts of the manifold, which will be located adjacent to the raft inflatable tube, shall be completely covered with a heavy layer of the same material as the patch to prevent chafing of the raft tube cloth.

3.3.2.1.3 Replacement. Replacment of any expendable parts shall be possible from the ends of the manifold body. The removable end plugs of the manifold body shall be constructed of the aluminum alloy conforming to QQ-A-225/5, Temper T4, and shall be anodized in accordance with MIL-A-8625. The end plugs shall conform to Drawing 63A120H1-55.

3.3.2.2 LRU-15/A and F-2B life rafts. The inlet manifold of the LRU-15/A and F-2B life rafts shall consist of a "Y" fitting with or without a bleeder valve, as applicable, and two inlet valves with or without valve checks, as applicable. The "Y" manifold shall substantially distribute the carbon dioxide charge equally through the inlet valves into each flotation tube. The "Y" manifold, when attached to a life raft, shall not interfere with the attachment or removal of a charged carbon dioxide cylinder and valve. The inlet manifold assembly for the LRU-15/A life rafts shall conform to Drawing 63A120H1-25 or -26, as applicable. The inlet manifold assembly for the F-2B raft shall conform to Drawing 63A120H1-25.

3.3.2.2.1 "Y" fitting.

3.3.2.2.1.1 Without air vent. The "Y" manifold fitting assembly, without the air vent, shall conform to Drawing 63A120H1-37 and is to be used on LRU-15/A rafts which are to be dropped to survivors. This manifold shall not be installed on LRU-15/A rafts which are to be stowed in the wing of an aircraft, or on any F-2B rafts.

3.3.2.2.1.2 With air vent. The "Y" manifold fitting assembly to be installed on LRU-15/A rafts which are to be installed in the wing of an aircraft, and all F-2B rafts shall be in accordance with Drawing 63A120H1-25. When installed on the life raft, the vent valve end shall point away from the carbon dioxide cylinder and shall be in the open position.

3.3.2.2.2 Inlet valves. The inlet valves shall conform to MIL-V-8883, Type II. Each "Y" fitting shall contain two inlet valves. The inlet valves used with the air vent "Y" manifold shall not contain any valve checks.

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3.3.3 Carbon dioxide cylinder.

3.3.3.1 Applicable to the Army and Navy. The empty carbon dioxide cylinder shall conform to MIL-C-7905 and MS26545. Each carbon dioxide cylinder shall contain a syphon tube conforming to Drawing 63A120H1. Each carbon dioxide cylinder shall be charged with carbon dioxide conforming to BB-C-101, Grade B, Type I. The carbon dioxide cylinders, the length of the syphon tube, and the carbon dioxide charge required for the various life rafts shall conform to Drawing 63A120H1.

3.3.3.2 Applicable to the Air Force. The empty carbon dioxide cylinder shall conform to MIL-C-7905 and MS26545. Each carbon dioxide cylinder shall contain a syphon tube conforming to Drawing 55C3689. Each carbon dioxide cylinder shall be charged with carbon dioxide conforming to BB-C-101, Grade B, Type I, and nitrogen conforming to BB-N-411, Type I, Class 1, Grade A. The carbon dioxide cylinders, the length of the syphon tube, and the carbon dioxide and nitrogen charge required for the various life rafts shall conform to Drawing 55C3689.

3.3.4 Snaphook or Looped End Cable Assemblies

3.3.4.1 Applicable to the Army and Navy. The cable assemblies shall conform to the requirements of Drawinbgs 63A120H1, 1106AS103, and, 1106AS104, as applicable.

3.3.4.2 Applicable to the Air Force. The cable assemblies shall conform to the requirements of Drawing 55C3689.

3.3.5 Pull Cable Housings

3.3.5.1 Applicable to the Army and Navy. The pull cable housings shall conform to the requirements of 63A120H1 and 1106AS102 as applicable

3.3.5.2 Applicable to the Air Force. The pull cable housings shall conform to the requirements of 55C3689.

3.3.6 Safety wiring.

3.3.6.1 Applicable to the Army and Navy. A safety wire and warning tag shall be attached to the inflation valve assembly in accordance with the requirements of Drawing 63A120H1. A red tag shall be attached to the safety wire and shall contain the following information:

WARNING

WEIGH CYLINDER IMMEDIATELY BEFORE INSTALLING ON LIFE RAFT. DO NOT INSTALL IMPROPERLY CHARGED CYLINDER OR IMPROPERLY SAFETIED OR UNSAFETIED INFLATION GEAR ASSEMBLY.

3.3.6.2 Applicable to the Air Force. The inflation equipment for the Air Force shall be safety wired in accordance with Drawing 55C3689.

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The safety wire shall be installed prior to shipment. The inflation equipment installed on the life rafts shall also be safety wired, as aforementioned, prior to shipment of the life rafts. A red warning tag, containing the following information, shall be attached to the safety wire:

WARNING
WEIGH CYLINDER IMMEDIATELY BEFORE INSTALLING ON LIFE RAFT. DO NOT INSTALL IMPROPERLY CHARGED CYLINDER OR IMPROPERLY SAFETIED OR UNSAFETIED INFLATION GEAR ASSEMBLY

3.3.7 Screw threads. All the screw threads shall conform to FED-STD-H28 and Drawing 63A120H1.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The applicable syphon tube and carbon dioxide cylinder shall be interchangeable with any valve regardless of the manufacturer.

3.5 Weight. When determined as specified in 4.6.2, the weight of the various components of the inflation equipment shall conform to Table I.

3.6 Performance characteristics.

3.6.1 Cylinder valve.

3.6.1.1 Leakage. When inspected as specified in 4.6.3.1 through 4.6.3.1.3, the valve shall not leak when exposed at the following temperatures: 70 ± 2°F, -65 ± 2°F and 160 ± 2°F.

3.6.1.2 Hydrostatic pressure of the frangible disc. The frangible disc, when the valve is exposed at a temperature of 160 ± 2°F, shall burst at a pressure between 3150 and 3500 pounds per square inch gage (psig) without any damage to the valve, when inspected as specified in 4.6.3.2.

3.6.2 Actuator assemblies.

3.6.2.1 Applicable to the Army and Navy.

3.6.2.1.1 Snaphook or Looped End Cable Assemblies. When inspected and tested as specified in 4.6.4.1.1, the snaphook or looped end cable assemblies shall conform to the visual, dimensional, and test requirements of Drawing 1106AS103 or 1106AS104, as applicable.

3.6.2.1.2 Pull Cable Housing. When inspected and tested as specified in 4.6.4.1.2, the pull cable housing shall conform to the visual, dimensional, and tests requirements of Drawing 1106AS102.

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3.6.2.2 Applicable to the Air Force.

3.6.2.2.1 Strength of Attachments to the Pull Cable. After assembly of the pull cable the ball, snaphook, socket and metal tube (loop), shall withstand a pull of 100 pounds, when inspected as specified in 4.6.4.2.1.

3.6.3 LRU-1/P, LRU-12/A, LRU-13/A, LRU-14/A life rafts inlet manifold.

3.6.3.1 Patch adhesion.

3.6.3.1.1 As received. The molded rubber patch in no instance shall strip away from the manifold and the rubber patch shall not fail below 75 pounds, when inspected as specified in 4.6.5.1.1.

3.6.3.1.2 After aging. The molded rubber patch in no instance shall strip away from the manifold and the rubber patch shall not fail below 65 pounds, when inspected as specified in 4.6.5.1.2.

3.6.3.2 Back pressure leakage. The inlet manifold shall not have any back pressure leakage, when inspected as specified in 4.6.5.2.

3.6.3.3 Intermittent operation. The inlet manifold shall not malfunction or develop structural defects, when a pressure of 60 psig is applied intermittently as specified in 4.6.5.3.

3.6.4 Assembled inflation equipment.

3.6.4.1 Coupling nut torque. When inspected as specified in 4.6.6.1, the coupling nut of the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold hose and of the LRU-15/A and F-2B life raft "Y" fitting (with or without a bleeder valve) and the inlet valves shall withstand a tightening torque of 30 foot pounds without failure, resultant leakage, or permanent deformation to the coupling nut, inlet manifold hose, inlet manifold, valve, "Y" fitting, or inlet valves, as applicable.

3.6.4.2 Operating force. When inspected as specified in 4.6.6.2, the safety wire shall break, the pull cable shall pull free of the housing, and the valve shall not unseat until a pull force of 12 $\pm$ 3 pounds is applied. The valve shall fully unseat within the previously stated range.

3.6.4.3 Operation. When inspected as specified in 4.6.6.3 through 4.6.6.3.2, as applicable, the inflation assembly shall discharge at least 90 percent of the carbon dioxide or carbon dioxide plus nitrogen content of cylinder, as applicable, in the time and temperature specified in Table II. The gas shall discharge through all the holes of each inlet diffuser or inlet valve, as applicable, without any indication of leakage.

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The carbon dioxide shall pass into the atmosphere only through the inlet diffuser or inlet valve holes, as applicable, and not at any other place. The inlet manifold or "Y" fitting, as applicable, shall discharge the carbon dioxide equally through the inlet diffusers or inlet valves, as applicable. The vent valve of the "Y" fitting, if used, shall close when the carbon dioxide enters the "Y" fitting. The weight of the charge shall conform to Drawing 63A120H1 or 55C3689, as applicable.

3.6.4.4 Vibration. The loss in weight of the carbon dioxide or carbon dioxide plus nitrogen charge shall not exceed one percent (based on the initial weight), when the assembled inflation equipment is vibrated and stored as specified in 4.6.6.4. The assembled inflation equipment shall not develop any leaks or structural defects. The cylinder valve shall then be inspected and conform to the requirements for leakage at 70°F, 3.6.1.1, and the assembled inflation equipment shall be inspected and conform to the requirements of the operating force, 3.6.4.2. The vibration inspection shall be performed in sequence with the endurance inspection. The vibration shall be performed first and when completed, the inflation equipment shall be inspected for conformance to the endurance inspection, 3.6.4.5.

3.6.4.5 Endurance. The assembled inflation equipment, when subject to 50 chargings and 50 dischargings as specified in 4.6.6.5, shall not develop any leaks or structural defects or show any evidence of malfunction or excessive wear. There shall be no difficulty in charging the carbon dioxide cylinder while attached to the inflation equipment. The cylinder valve shall then be inspected and conform to the requirements for leakage at 70°F, 3.6.1.1, and the assembled inflation equipment shall be inspected and conform to the requirements for operating force, 3.6.4.2.

3.6.4.6 Hydrostatic pressure.

3.6.4.6.1 At 3500 psig. When inspected as specified in 4.6.6.6.1, the assembled inflation equipment shall not deform, burst, or leak, when subjected to an internal hydrostatic pressure of 3500 psig.

3.6.4.6.2 At 5000 psig. When inspected as specified in 4.6.6.6.2, the assembled inflation equipment shall not deform, burst, or leak, when subjected to an internal hydrostatic pressure of 5000 psig.

3.6.4.7 Salt spray. When inspected as specified in 4.6.6.7, the assembled inflation equipment shall show no evidence of corrosion or damage. The assembled inflation equipment shall then be inspected and conform to the requirements for operating force, 3.6.4.2, and operation at 70°F, 3.6.4.3.

3.6.4.8 Drop inspection (applicable only to the Air Force). When inspected as specified in 4.6.6.8, the inflation equipment shall not be damaged or become deformed and shall function properly.

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3.7 Identification of product. Except for the LRU-15/A and F-2B life raft inlet valves, the cylinder valve, inlet manifold, and "Y" fittings shall be legibly and permanently marked in accordance with MIL-STD-130. The LRU-15/A and F-2B life raft valves shall be marked as specified in MIL-V-8883.

3.7.1 Identification of carbon dioxide cylinder. In addition to the markings required by MIL-STD-130, applicable specifications and/or standards, the carbon dioxide cylinder shall be weighed and marked, in 1/2 inch black letters and numerals, with the following information:

INFLATION ASSEMBLY (LRU-12/A, LRU-13/A, LRU-14/A, LRU-15/A, LRU-1/P or F-2B, as applicable)
GOVERNMENT PART NUMBER (Refer to Drawing 63A120H1 or 55C3689, as applicable)
TARE WT: (Empty Cylinder, Valve, Cable and Cable Housing)
CO_2 WT: (Refer to Drawing 63A120H1 or 55C3689, as applicable)
GROSS WT: (Tare Wt Plus CO_2Wt)

In addition, mark in 1 inch red letters:

WARNING - COMPRESSED GAS - DO NOT DROP

3.8 Workmanship. The inflation equipment shall not be misaligned nor contain any rough surface, nick, burr, sharp edge, crack, embedded foreign matter, detached component, loose particle, or other flaw. Because of the emergency and life support use of this equipment, the importance of providing a product of uniform excellent quality cannot be overemphasized. The inflation equipment shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The inflation equipment shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.1 and 4.3 through 4.3.1.1).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots to the requirements set forth in this specification (see 4.4).

4.3 First article inspection. The first article inspection of the carbon dioxide inflation equipment shall consist of examinations and tests listed in Table III.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit three complete units of assembled inflation equipment. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1e). Approval of the first article samples or the waiving of the first article inspection does not waive the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1f).

4.3.1.1 First article information. Upon completion of the first article inspection, recommendations and comments pertinent for use in monitoring production shall be forwarded by the Government Activity responsible for conducting the program (see 6.2.1f) to the Contracting Officer. One of the approved first article inspection sample units will be returned to the manufacturer for use in monitoring production. The other two units will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract.

4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following, as applicable:

Visual examination
Dimensional check
Weight
Leakage of the cylinder valve at 70°F
Hydrostatic pressure of the frangible disc

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Strength of the attachment of the ball, snap hook, socket, and metal tube (loop) to the pull cable
Patch adhesion (as received) of the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14A life rafts inlet manifold
Back pressure leakage of the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold
Coupling nut torque
Operating force
Operation at 70°F
Packaging
Snaphook or Looped End Cable
Pull Cable Housing

4.4.1 Sampling.

4.4.1.1 Inspection lot.

4.4.1.1.1 Components. An inspection lot size shall be expressed in units of one individual component of the inflation equipment, made essentially under the same conditions and from the same materials and constituents. The sample unit shall be one individual component of the inflation equipment.

4.4.1.1.2 Assembled inflation equipment. An inspection lot size shall be expressed in units of one assembled inflation equipment of one kind assembled essentially under the same conditions and from the same components. The sample unit shall be one assembled inflation equipment.

4.4.1.1.3 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing assembled inflation equipment or individual components, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing assembled inflation equipment or individual components, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for examinations and tests of the individual components and the assembled inflation equipment. The sample size, acceptance criteria, examinations, and tests required for the individual components and the assembled inflation equipment shall be as specified in Table IV.

4.5 Inspection conditions.

4.5.1 Atmospheric conditions. Unless otherwise specified, all inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 ± 18°F (25 ± 10°C), and a relative humidity of 80 percent or less. When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument reading.

4.6 Inspection methods.

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4.6.1 Visual examination.

4.6.1.1 Components and assembled inflation equipment. Every individual component or each component of the assembled inflation equipment, as applicable, shall be examined visually for critical defects to determine conformance to this specification. Each individual component or each component of the assembled inflation equipment, as applicable, selected as a sample unit from the lot shall be thoroughly checked dimensionally and examined visually for minor defects to determine conformance to this specification. The classification and list of defects, Tables V and VI, as applicable, shall be used to classify and enumerate the defects found.

4.6.1.2 Packaging. Each of the fully prepared shipping containers, containing individual components or assembled inflation equipment of one kind, as applicable, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, preservation, packing, and marking conform to this specification. The list of defects, Table VII, shall be used to enumerate the defects found.

4.6.2 Weight. The weight of the individual component or assembled inflation equipment, as applicable, shall be determined on any scale or balance capable of weighing to the nearest 0.01 of a pound.

4.6.3 Cylinder valve.

4.6.3.1 Leakage.

4.6.3.1.1 At 70°F. An applicable empty carbon dioxide cylinder shall be attached to the cylinder valve. The cylinder shall be charged with the proper amount of carbon dioxide or carbon dioxide plus nitrogen charge specified in Drawings 63A120H1 or 55C3689, as applicable. There shall be no difficulty in charging the cylinder through the valve outlet. The valve and charged cylinder shall be weighed as specified in 4.6.2 and shall be immersed, in a horizontal position, under a static head of 2 feet of water, at a temperature of 70 ± 2°F (21 ± 1°C), for a minimum of six hours and a maximum of seven hours. A steady stream of bubbles is an indication of leakage. Upon completion of the immersion period, the valve and charged cylinder shall be removed from the water, wiped dry, and reweighed. Any loss in weight is an indication of leakage.

4.6.3.1.2 At -65 °F. The aforementioned shall be repeated, except that the valve and charged cylinder shall be conditioned at -65 ± 2°F (-54 ± 1°C) for a minimum of six hours and a maximum of seven hours.

4.6.3.1.3 At 160°F. The aforementioned shall be repeated, except that the valve and charged cylinder shall be conditioned at 160 ± 2°F (71 ± 1°C) for a minimum of six hours and a maximum of seven hours.

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4.6.3.2 Hydrostatic pressure of the frangible disc. The valve and disc shall conform to the requirements of 3.6.1.2 when inspected as follows: The valve with an empty carbon dioxide cylinder attached and the hydraulic pressure medium shall be immersed under a static head of 2 feet of water, at a temperature of 160 ± 2°F (71 ± 1°C), for a minimum of one hour. At the end of one hour, hydraulic pressure shall be applied to the disc through the valve outlet at the rate of increase of approximately 100 psig per second until the disc ruptures.

4.6.4 Actuator assemblies.

4.6.4.1 Applicable to the Army and Navy.

4.6.4.1.1 Snaphook or Looped End Cable Asemblies. The cable assemblies shall be tested and examined in accordance with requirements of Drawings 63A120H1 and 1106AS103 or 1106AS104, as apllicable. The assemblies shall conform to 3.6.2.1.1."

4.6.4.1.2 Pull Cable Housing. The pull cable housing shall be tested and examined in accordance with the requirements of Drawings 63A120H1 and 1106AS102 for conformance to 3.6.2.1.2."

4.6.4.2 Applicable to the Air-Force.

4.6.4.2.1 Strength of the attachments to the pull cable. The ball shall be mounted in a suitably shaped jig having a hole of the proper size to clear the pull cable, but capable of holding the ball. The jig shall be clamped in one jaw and the cable secured in the other jaw of a suitable inspection apparatus. The speed of the movable jaw of the inspection apparatus under no load shall be 12 ± 1/2 inches per minute. A metal rod shall be inserted through the eye of the snap hook or through the loop of the pull cable, as applicable. The metal rod shall be attached to one jaw and the pull cable to the other jaw of the aforementioned inspection apparatus. The jaws shall be separated until failure occurs and shall pass the requirements of 3.6.2.2.1.

4.6.5 LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold.

4.6.5.1 Patch adhesion.

4.6.5.1.1 As received. The adhesion of the molded rubber patch to the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold shall be inspected for conformance to the requirements of 3.6.3.1.1 as follows. Five tabs, each one inch wide, shall be completely cut, on both sides of the patch, to the metal portion of the manifold. Two diametrically opposite tabs shall be clamped in the jaws of a suitable inspection apparatus with the movable jaw separating at a rate of 2 ± 1/8 inches per minute under no load. The jaws shall be separated until failure occurs. The inspection shall be repeated for each remaining four diametrically opposite tabs.

4.6.5.1.2 After aging. The inlet manifold shall be exposed in an atmosphere of 170 ± 2°F (77 ± 1°C) for 168 ± 1/4 hours. The inlet manifold shall then be removed and exposed to a temperature of 75 ± 5°F (24 ± 2°C).

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After a minimum of 16 hours and not more than 96 hours exposure at 75 ± 5°F, the patch adhesion after aging shall be inspected for conformance to the requirements of 3.6.3.1.2 as specified in 4.6.5.1.1.

4.6.5.2 Back pressure leakage. The LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold shall be submerged in water. While submerged, the two inlet diffusers shall be pressurized simultaneously to 5.0 ± 0.1 psig with oil and liquid water free air. The manifold shall then be examined for conformance to the requirements of 3.6.3.2.

4.6.5.3 Intermittent operation. The LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold shall be subjected to the following pressure cycles:

> Sixty cycles of 0-60-0 psig (approximately 10 cycles per hour). The peak pressure shall be reached in not more than 15 seconds and shall be held for 5 minutes and reduced to zero in not more than 15 seconds. A period of 60 seconds under no pressure shall elapse before the start of the next cycle. Oil and liquid water free air shall be used and shall be applied to the inlet side of the inlet manifold. Upon completion of the pressure cycles, the inlet manifold shall be examined for conformance to the requirements of 3.6.3.3. The inlet manifold shall then be examined for any malfunction during the operation inspection, 4.6.6.3 through 4.6.6.3.2.

4.6.6 Assembled inflation equipment.

4.6.6.1 Coupling nut torque. A torque of 30 foot-pounds shall be applied to each coupling nut of the inlet manifold hose for attachment to the valve and to the LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold or for the attachment of the LRU-15/A and F-28 life raft "Y" fitting (with or without bleeder valve) to the valve and to the inlet valves, as applicable. The coupling nuts, inlet manifold hose, inlet manifold, valve, "Y" fitting, or inlet valves, as applicable, shall be examined for signs of failure and permanent deformation and shall conform to the requirements of 3.6.4.1. The connections shall be examined for leakage during the operation inspection, 4.6.6.3 through 4.6.6.3.2, as applicable.

4.6.6.2 Operating force. This test shall be performed at 70 ± 2°F (21 ± 1°C). The specified assembly, charged and safety wired in accordance with the requirements of Drawings 63A120H1 or 55C3689, as applicable, shall be securely restrained in a horizontal position in a cylinder vise. A calibrated tension measuring device shall be attached to the snaphook or looped end of the pull cable assembly and a pull force slowly applied in line with the centerline of the cylinder. When tested in this manner, the assembly shall conform to the requirements of 3.6.4.2.

4.6.6.3 Operation.

4.6.6.3.1 At 70°F. The assembled inflation equipment charged and safety wired in accordance with Drawings 63A120H1 or 55C3689, as applicable, shall be conditioned at 70 ± 2°F (21 ± 1°C) for a minimum of 12 hours.

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The assembled inflation equipment shall be weighed on a scale or balance to the nearest 0.01 of a pound. The "Y" fitting bleeder valve, if used, shall be in the open position. The carbon dioxide filled cylinder shall be held in a horizontal position and actuated. At the end of the time specified in 3.6.4.3 for the assembled inflation equipment being inspected, the time being recorded from the instant the carbon dioxide filled cylinder is actuated, the valve shall be returned to the closed position, wiped dry, and the assembly reweighed. Ninety percent of the carbon dioxide or carbon dioxide plus nitrogen shall have been discharged. The carbon dioxide or carbon dioxide plus nitrogen shall have passed into the atmosphere through all the holes of each inlet diffuser or through all the holes of each inlet valve, as applicable, and not through any other part of the assembly. The "Y" fitting bleeder valve, if used, shall have closed. Upon completion of the aforementioned, the valve shall be opened and the reminder of the valve shall be discharged from the cylinder. The assembled inflation equipment shall then be wiped dry, reweighed, and the total amount of the charge in the cylinder determined.

4.6.6.3.2. At - 40°F. The assembled inflation equipment shall be inspected as specified in 4.6.6.3.1, except that the inflation equipment shall be conditioned and inspected at -40° ± 2° (-40 ± 1°C).

4.6.6.4 Vibration. The assembled inflation equipment charged and safety wired in acordance with Drawings 63A120H1 or 55C3689, as applicable, shall be weighed (initial weight) on a scale or balance to the nearest 0.01 of a pound and then vibrated for 18 hours at an amplitude of 6G ± 10 percent and at a frequency of 55 Hz ± 1 Hz. The inflation equipment shall be vibrated in three mutually perpendicular axes, six hours in each direction and shall be hard mounted to the vibration fixture. The input control sensing device shall be rigidly attached to the fixture adjacent to the attachment points of the test item. The inflation assembly shall then be reweighed and stored at 70 ± 2°F (21 ± 1°C) for 24 days. The assembled inflation equipment shall be weighed daily during the storage period. Upon completion of the 24 day storage period, the inflation assembly shall be vibrated for 18 hours as aforementioned and then weighed. The loss in weight of the carbon dioxide charge shall not exceed that specified in 3.6.4.4. The cylinder valve shall then be subjected and conform to the leakage inspection at 70°F, 4.6.3.1.1, and the assembled inflation equipment shall be subjected and conform to the operating force inspection, 4.6.6.2. Upon completion of the aforementioned, the assembled inflation equipment shall be subjected and conform to the endurence inspection, 4.6.6.5.

4.6.6.5 Endurance. The inflation assembly shall meet the endurence requirements of 3.6.4.5 when inspected as follows: The carbon dioxide cylinder, while attached to the assembled inflation equipment, shall be charged with the applicable quantity of carbon dioxide or carbon dioxide plus nitrogen (reference Drawings 63A120H1 or 55C3689 and then discharged at 70° ± 2°F (21 ± 1°C) for 50 complete cycles. The cylinder shall be charged thru the valve outlet without any difficulty.

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No component of the assembled inflation equipment shall show any evidence of malfunction or excessive wear. Upon completion of the 50 cycles, the cylinder valve shall be subjected and conform to the leakage inspection at 70°F, 4.6.3.1.1, and the assembled inflation equipment shall be subjected and conform to the operating force inspection, 4.6.6.2.

4.6.6.6. Hydrostatic pressure.

4.6.6.6.1. At 3500 psig. The frangible disc of the valve shall be removed and replaced by a suitable plug. The valve shall be unseated and shall not contain the carbon dioxide cylinder. The LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold or the "Y" fitting (with or without bleeder valve), as applicable, shall be attached to the valve. The inlet valves shall not be attached to the "Y" fitting. The openings in the inlet diffuser and the outlets of the "Y" fitting, as applicable, shall be suitable plugged. The bleeder valve of the "Y" fitting, if used, shall be in the closed position. A hydrostatic preesure of 3500 psig shall be applied thru the valve inlet (carbon dioxide cylinder connection) for 5 minutes. The pressure shall be increased from 0.0 ,psig to 3500 psig in 35 $\pm$ 3 seconds. At the end of 5 minutes and while under pressure, the inflation equipment shall be examined for conformance to the requirements of 3.6.4.6.1. Upon completion of the aforementioned, the pressure shall be released, the inlet manifold or "Y" fitting, as applicable, shall be disconnected from the valve and the valve shall be seated. Pressure shall be applied and the valve examined as aforesaid.

4.6.6.6.2 At 5000 psig. The frangible disc of the valve shall be removed and replaced by a suitable plug. The valve shall be unseated and shall not contain the carbon dioxide cylinder. The LRU-1/P, LRU-12/A, LRU-13/A and LRU-14/A life rafts inlet manifold or the "Y" fitting (with or without the bleeder valve), as applicable, shall be attached to the valve. The inlet valves shall not be attached to the "Y" fitting. The openings in the inlet diffuser and the outlets of the "Y" fitting, as applicable, shall be suitably plugged. The bleeder valve of the "Y" fitting, if used, shall be in the closed position. A hydrostatic pressure of 5000 psig shall be applied through the valve inlet (carbon dioxide cylinder connection) for 5 minutes. The pressure shall be increased from 0.0 psig to 5000 psig in 50 $\pm$ 3 seconds. At the end of 5 minutes and while under pressure, the inflation equipment shall be examined for conformance to the requirements of 3.6.4.6.2. The pressure shall be released and the inflation equipment assembly again examined for conformance to the requirements of 3.6.4.6.2.

4.6.6.7 Salt spray. The assembled inflation equipment charged and safety wired in accordance with Drawings 63A120H1 or 55C3689, as applicable, shall be exposed to salt spray for 100 hours in accordance with FED-STD-151, Method 811. The assembled inflation equipment shall be exposed in a horizontal position, simulating the position as when attached to the applicable life raft. Suitable provisions shall be made to prevent the entry of spray into the openings of the inlet diffusers or the discharge holes of the inlet valves, as applicable. At the expiration of the salt spray exposure, the assembled inflation equipment shall be examined for conformance to the requirements of 3.6.4.7.

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The assembled inflation equipment shall then be subjected and conform to the operation inspection at 70°F, 4.6.6.3.1, and the operating force inspection, 4.6.6.2.

4.6.6.8 Drop inspection (applicable only to the Air Force). The assembled inflation equipment shall be installed on the applicable life raft. The life raft shall be folded and inserted into its carrying case as specified in the applicable life raft specification. The life raft in its carrying case shall be dropped onto water from an aircraft flying at approximately 100 miles per hour at an altitude of approximately 100 feet. The life raft shall be removed from the water and inflated in accordance with the instructions on the carrying case. The life raft shall inflate in the time specified in the applicable life raft specification and the pressure in the tubes shall also comply with the applicable life raft specification. The inflation equipment shall then be examined for conformance to the requirements of 3.6.4.8.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified (see 6.2.1).

5.1.1 Level A. Each assembled inflation equipment shall be preserved and cushioned in accordance with MIL-P-116, Method 1C-3. No contact preservation is required. Each preserved assembled inflation equipment shall be packaged within a snug fitting fiberboard container conforming to PPP-B-636, Type CF, Weather Resistant Class, Variety SW, Grade V3c. Closure and sealing of the containers shall be in accordance with the appendix to PPP-B-636. All the seams and joints shall be sealed with a minimum of 3 inch wide tape conforming to PPP-T-60, Type III, Class 1. If the pull cable assembly, manifold, valve, and valve with charged cylinder are purchased as individual items, each shall be packaged in the same manner as the assembled inflation equipment.

5.1.2 Level C. The assembled inflation equipment or the individual component of the inflation equipment, as applicable, shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2.1). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight. Each shipping container shall contain the same part number (see Drawings 63A120H1 and 55C3689, as applicable).

5.2.1 Level A. The assembled inflation equipment or the individual component of the inflation equipment, as applicable, packaged as specified in 5.1.1, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Weather Resistant Class, Variety SW, Grade W5c or W5s. Each container shall be constructed, closed, and reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

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5.2.2 Level B. The assembled inflation equipment or the individual component of the inflation equipment, as applicable, packaged as specified in 5.1.1, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. The packaged assembled inflation equipment or the individual component of the inflation equipment, as applicable, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall ensure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year). The exterior containers shall also be marked to conform to the rules and regulations applicable to the mode of transportation. In addition and when applicable, regardless of the mode of transportation, the carbon dioxide filled cylinder, the interior container, and the exterior shipping container shall contain a green label in accordance with the Code of Federal Regulations, Title 49, Section 100-199, for the Transportation of Explosives and Other Dangerous Articles. In addition, the above shall be marked "NON-FLAMMABLE COMPRESSED GAS" and "CARBON DIOXIDE LIQUIFIED". The size and color of the markings shall conform to the Code of Federal Regulations for the Transportation of Explosives and Other Dangerous Articles. The markings shall be conspicuous and legible.

6. NOTES

6.1 Intended use. The inflation equipment covered by this specification is intended for use as the carbon dioxide inflation system for the various multiplace life rafts.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, Number and Date of this specification, including any amendments

b. Applicable Government Part number and National Stock Number

c. Applicable Government Drawings, including revisions

d. Quantity desired

e. Whether First Article inspection is required (see 4.3.1).

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f. Name and address of the First Article inspection facility and the name and address of the Government activity responsible for conducting the first article inspection program (see 4.3.1. and 4.3.1.1).

g. Selection of applicable levels of preservation, packaging and packing. (see 5.1 and 5.2).

h. Items of data required (see 6.2.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.
4.3.1.1	Inspection and test reports	DI-T-5329

(Copies of data item descriptions required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, it shall be tested under the appropriate provision of 7-104.55 of the Defense Acquisition Regulation. The first article should be a preproduction sample. The contracting officer should include specific instructions in acquisition [illegible] first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS

(Project 4220-0287)

Review activities:
Army - ME
Navy - CG, SH
Air Force - 22, 84

User activities:
Navy - MC

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TABLE I. Weight of the components of the inflation equipment.

ITEM	PART (DRAWING) NUMBER	MAXIMUM WEIGHT
Cylinder Valve	63A129H1-19	2.0 pounds
Inlet manifold	63A120H1-30 (8 ± 1/4 inches)	16.0 ounces
Air vent "Y" fitting	63A120H1-25 (Y-Portion only)	14.0 ounces
"Y" fitting (without air vent)	63A120H1-37	6.0 ounces

TABLE II. Discharge time and temperature.

ASSEMBLY PART (DRAWING) NUMBER	RAFT DESIGNATION	MAXIMUM DISCHARGE TIME (seconds)	
		At 70°F	At -40°F
63A120H1-15	LRU-14/A	9	21
63A120H1-13	LRU-12/A	6	15
63A120H1-14	LRU-13/A	9	21
63A120H1-16, -17, -18	LRU-15/A	20	40
55C3689-1 and 63A120H1-30	LRU-1/P	9	21
55C3689-2 and 63A120h1-30	-------	17	40
55C3689-3 and 63A120H1-30	-------	6	15
55C3689-4 and 63A120H1-25	F-2B	20	40

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TABLE III First Article Test and Examinations

INSPECTION	REQUIREMENT PARAGRAPH	TEST AND EXAMINATION PARAGRAPH
Visual Examination	-----	4.6.1
Dimensional Check	------	4.6.1.1
Weight	3.5	4.6.2
Leakage	3.6.1.1	4.6.3.1
Hydrostatic Pressure of the Frangible Disc	3.6.1.2	4.6.3.2
Actuator Assemblies	3.6.2.	4.6.4
Patch Adhesion	3.6.3.1	4.6.5.1
Back Pressure Leakage	3.6.3.2	4.6.5.2
Intermittent Operation	3.6.3.3	4.6.5.3
Coupling Nut Torque	3.6.4.1	4.6.6.1
Operating Force	3.6.4.2	4.6.6.2
Operation	3.6.4.3	4.6.6.3
Vibration	3.6.4.4	4.6.6.4
Endurance	3.6.4.5	4.6.6.5
Hydrostatic Pressure	3.6.4.6	4.6.6.6
Salt Spray	3.6.4.7	4.6.6.7
Drop Inspection	3.6.4.8	4.6.6.8

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TABLE IV Sample size, acceptance criteria, examinations and tests of the individual components and the assembled inflation equipment.

Inspection	Paragraph		Sample Size	Acceptance Criteria
	Requirement	Method		
Visual examination	----	4.6.1.1	Every individual component or assembled inflation equipment, as applicable, for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defect and an Acceptance Quality Level of 6.5 defects per 100 units for minor defects.
Dimensional check	---	4.6.1.1	Inspection Level S-3	An Acceptable Quality Level of 10.0 defects per 100 units.
Weight	3.5	4.6.2	Inspection Level I 2/	Acceptance number zero, rejection number 1
Leakage of cylinder valve at 70°	3.6.1.1	4.6.3.1.1	Every cylinder valve 1/	Reject all defective units
Hydrostatic pressure of the frangible disc	3.6.1.2	4.6.3.2	Inspection Level S-1 1/ 2/	Acceptance number zero, rejection number 1
Strength of attachements to the pull cable (applicable to the Air Force)	3.6.2.2.1	4.6.4.2.1	Inspection Level S-1 1/ 2/	Acceptance number zero, rejection number 1
Snaphook or Looped End Cable Assemblies	3.6.2.1.1	4.6.4.1.1	Inspection Level S-1 1/ 2/	Acceptance Number 0 Rejection Number 1
Pull Cable Housing	3.6.2.1.2	4.6.4.1.2	Inspection Level S-1/ 2/	Acceptance Number 0 Rejection Number 1

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TABLE IV Sample size, acceptance criteria, examinations and tests of the individual components and the assembled inflation equipment.-Continued

Inspection	Paragraph		Sample Size	Acceptance Criteria
	Requirement	Method		
Patch adhesion (as received) of the life rafts inlet manifold 4/	3.6.3.1.1	4.6.5.1.1	Inspection Level S-1 1/ 2/	Acceptance number zero, rejection number 1
Back pressure leakage of the life rafts inlet manifold 4/	3.6.3.2	4.6.5.2	Every LRU-1/P, LRU-12/A, LRU-13/A, LRU-14/A life rafts inlet manifold 1/	Reject all defective units
coupling nut torgue 3/	3.6.4.1	4.6.6.1	Every inflation assembly	Reject all defective units
Operating force	3.6.4.2	4.6.6.2	Every inflation assembly	Reject all defective units
Operation at 70°F	3.6.4.3	4.6.6.3.1	Every inflation assembly	Reject all defective units
Packaging	Section 5	4.6.1.2	Inspection level S-2	An Acceptable Quality Level of 4.0 percent defective

1/ These inspections shall also apply to, when these items are components of the assembled inflation equipment.

2/ The sample size shall be based only on the applicable sample size code letter correspondinng to the specified inspection level of MIL-STD-105.

3/ Upon completion of the coupling torgue, the operation at 70°F shall be conducted only.

4/ These tests are applicable to the inlet manifold of the LRU-1/P, LRU-12/A, LRU-13/A, LRU-14/A life rafts.

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TABLE V Classification of the defects for visual examination of the components and the assembled inflation equipment.

CRITICAL	MINOR
1. Any surface rough, misaligned or contains nicks, sharp edges, cracks burrs, embedded foreign matter, loose particles, or other flaws. 2. Any finish missing or any component imrpoperly finished. 3. Any component missing, malformed, corroded, fractured or otherwise damaged. 4. Any component loose, detached, nor as specified; or otherwise not securely retained. 5. When applicable, not safety wired as specified; material of the safety wire not as specified. 6. When applicable, leaky carbon dioxide filled cylinder. 7. When applicable, any snap fastener mismated or improperly clinched resulting in cutting of the cloth.	201. Any identification or red warning tag marking missing, incorrect, illegible, or incomplete. 202. Red warning tag missing.

TABLE VI List of defects for finished dimensions of the individual components of the assembled inflation equipment.

EXAMINE	DEFECT
Measure the individual component or each component of the assembled inflation equipment, as applicable.	Any measurement deviating from the dimensions and tolerance as specified in Drawings 63A120H1 and 55C3689, as applicable, shall be enumerated as a dimensional defect.

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TABLE VII List of defects for packaging.

ITEM	DEFECTS
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exteriror containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; more than one kind in the same container.

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